

TASK 4: Method Design & Modular Calculator

Tools:

- Primary: IntelliJ IDEA
- Alternatives: Eclipse

Hints / Mini Guide:

1. Create a calculator application where each operation is a separate method.
2. Demonstrate method overloading using different parameter types.
3. Implement reusable utility methods.
4. Use return values effectively.
5. Show pass-by-value behavior in Java.
6. Add exception handling for divide-by-zero.
7. Structure code to reduce redundancy.
8. Test each method independently.

Deliverables:

- Modular calculator program
- Method documentation comments

Interview Questions Related To Above Task:

- What is method overloading?
- How does Java pass parameters?
- Difference between return and void?
- What is stack memory?
- Why modular code is important?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

